Is the Government Budget Sustainable? $r < g$ for Intermediate Macroeconomics

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Abstract

I present a method for introducing Intermediate Macroeconomics students to the contemporary macroeconomic question of government budget sustainability under $r < g$, that is, when the interest rate on government debt is less than the growth rate of nominal GDP. The theory extends the standard intertemporal budget constraint and admits a simple diagrammatic representation. My analysis proceeds in two steps: a simple model with a fixed interest rate and a richer version where the interest rate increases with government debt. If the interest rate is fixed, the government can sustain any deficit as long as $r < g$, but if the interest rate is endogenous, the government faces clear constraints. I take the model to the data with a clear parameterization. According to the model, the current government budget for the US (and most rich countries) is unsustainable.

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The government budget is a central issue in macroeconomics, and students are naturally interested in the topic. Whether the government can sustain its current fiscal position is a pressing question. Students in Intermediate Macroeconomics may assume that the government budget is only sustainable if tax revenue exceeds government expenditure. Clearly, the US government spends more than it collects in tax revenue, so the budget appears unsustainable.

This analysis ignores two key variables: the nominal interest rate on government debt, r, and the growth rate of nominal GDP, g. Blanchard (2019) points out that if the growth rate of the economy exceeds the interest rate on government debt, then, in theory, the government can borrow without limit. In other words, if $r < g$, then there is potential for "free lunch." Unsurprisingly, $r < g$ has generated a fruitful debate.

Since $r < g$ usually holds in the US, that would seem to be good news. The US tends to maintain steady growth, and the US government can borrow at relatively low interest rates. US Treasury bonds are considered very safe assets, so investors demand relatively little interest, and the 21st-century global economy has been characterized by declining interest rates. However, Blanchard (2019) and the related literature caution that while the free lunch condition is a good starting point, it is too simplistic, and determining the government's fiscal space is a complex quantitative question.

In this paper, I propose a simple method for teaching the $r < g$ debate in Intermediate Macroeconomics. I begin with a theory with intuitive diagrams and steady-state reasoning. Except for an optional extension, it does not require calculus. Then, I take the model to the data using a straightforward parameterization. It turns out, like many questions in economics, that the answer to whether the US government budget is sustainable is "it depends."

My theory is a simple extension of the basic intertemporal government budget constraint. It explicitly builds on the equations in Jones (2024), but many intermediate macroeconomics textbooks provide easy jumping-off points. Then, I assume that the primary deficit-to-GDP ratio and the growth rate of nominal GDP are constant. Using a little algebra, I derive an expression for the evolution of the debt-to-GDP ratio over time. In the model, the government budget is considered unsustainable simply if the debt-to-GDP ratio is predicted to grow infinitely large. The model is conveniently summarized in a diagram of the change in the debt-to-GDP ratio, similar to the Solow growth model. In the first version of the model, I assume that the interest rate is constant. Using the diagram, I show that if $r < g$, then the debt-to-GDP ratio is guaranteed to converge to a steady state. Therefore, any deficit is sustainable, and the government can borrow without limits.

However, when I make the model more realistic, the analysis quickly becomes less optimistic. In the second version of the model, I assume that the interest rate on government debt is an increasing linear function of the debt-to-GDP ratio. After accounting for endogenous interest rates, the redrawn diagram shows that the government cannot sustainably run any deficit. In an optional extension with calculus, students can analytically solve for the maximum sustainable deficit-to-GDP ratio. Thus, given the current debt-to-GDP ratio, primary deficit-to-GDP ratio, growth rate of nominal GDP, and interest rate on government debt, students can determine whether a country's budget is sustainable.

Then, I demonstrate how to take the model to the data. My benchmark results are for the US in 2025. Almost every parameter in the model is easily identifiable in the data. The exception is the sensitivity of the interest rate to the debt-to-GDP ratio, which I borrow from other studies. I then use the parameters to compute the dynamic debt-to-GDP diagrams for the US. If the interest rate is constant, then since $r < g$, the model predicts that the current US government budget is sustainable. In particular, the model predicts that the debt-to-GDP ratio will eventually converge to approximately 200%, after which it will remain constant. However, if the interest rate is endogenous to the level of debt, then the current government budget is not sustainable, and the debt-to-GDP ratio is predicted to explode. The model suggests that the US must reduce its primary deficit by half to return to a sustainable budget.

Finally, I run the same analysis on a sample of other countries for 2025 and 2015. Due to the narrowing gap between growth and interest rates, many government budgets, including the US', flipped from sustainable to unsustainable between 2015 and 2025. I also include sample problems in the appendix.

Beyond students' natural curiosity about the subject, my method has several pedagogical benefits. The model relies on convergence to a steady state, a concept that students learn while studying the Solow growth model but rarely use again. A key determinant of the model's predictions is whether the interest rate is endogenous or exogenous, which would clarify the distinction between the two. Since almost every parameter in the model is easily observable, it is easy and rewarding to bring the model to the data. Budget sustainability reduces to a rigorous yet intuitive mathematical definition. Finally, the model emphasizes the relationship between stocks and flows.

The issue of government budget sustainability will be relevant for the foreseeable future. Due to the combination of Social Security, Medicare, and Medicaid, along with an aging population, the Congressional Budget Office projects that government debt and deficits will continue to increase (Congressional Budget Office, 2026). For a rigorous but accessible analysis of how the US can get on a sustainable path, students may enjoy reading Dynan and Elmendorf (2025).

The remainder of this paper is as follows. I contextualize the paper within the research and teaching literature in Section 1. Section 2 presents the theory. I parameterize and quantitatively analyze the model in Section 3 before concluding in Section 4. Sample questions are provided in Appendix A.

1 Related Literature

First, I review the research literature on r, g, and the sustainability of the government budget. The following papers may be useful for instructors who seek to understand the issue in greater depth.

Though not the first word on the topic, Blanchard (2019) is foundational. He argues that since interest rates are typically below the nominal growth rate of GDP, government debt is not as much a risk to sustainability as it is a risk to efficiency. Reis (2021) and Cao et al. (2024) argue that since public debt crowds out private capital, increases in government debt may decrease welfare and eventually push the interest rate above the growth rate. Ball et al. (1998) suggests that since $r < g$ is uncertain, government deficits are a gamble. The uncertainty of $r < g$ is also the focus of Mehrotra and Sergeyev (2021) andBarrett (2018). In addition to warning about uncertainty, Cochrane (2021) contends that the size of the gap between r and g is not large enough to significantly affect policymaking.

My analysis is most similar to Mian et al. (2025), particularly in how I parsimoniously endogenize the interest rate on government debt. Like Mian et al. (2025), my model predicts that current policy will eventually raise interest rates until the debt-to-GDP ratio explodes.

Next, I review the teaching literature. I reviewed nine intermediate macroeconomics textbooks (Abel et al., 2024; Blanchard, 2025; Carlin and Soskice, 2023; Gordon, 2012; Jones, 2024; Mankiw, 2025; Mishkin, 2015; Williamson, 2018). Every book discusses the government budget to some extent. My theory begins with the intertemporal government budget constraint in Jones (2024). Abel et al. (2024), Blanchard (2025), and Carlin and Soskice (2023) derive a dynamic equation for the debt-to-GDP ratio, which is how my extension begins.

My paper is most similar to Chapter 7 of Carlin and Soskice (2023). Specifically, Carlin and Soskice (2023) teach a model similar to my constant-r model. However, I go beyond by (1) proposing a model where r is endogenous and (2) producing quantitative results, both of which make the analysis more similar to "real " macroeconomics. My quantitative analysis also brings the model to life and equips students with the tools to evaluate any government's budget.

Other teaching literature on the government budget includes Gale (2020) and Aguilar and Soques (2015). Finally, this paper joins others that seek to more effectively bridge intermediate macroeconomics with modern macroeconomic research (Loewy, 2026; Hashimzade et al., 2026; Casey, 2026; Neumuller et al., 2018; Solis-Garcia, 2018; Bhattacharya et al., 2018; Poutineau and Vermandel, 2015; Buttet and Roy, 2014; Ramamurthy and Sedgley, 2013; Bofinger et al., 2009; Kwok, 2007; Brevik and Gärtner, 2007; Bofinger et al., 2006;

Walsh, 2002).

2 Theory

2.1 Dynamics of Debt-to-GDP Ratio

A dynamic equation for the debt-to-GDP ratio summarizes the model. For convenience, I will derive the equation starting from the standard government budget constraint as written in Jones (2024).

The government enters time t with a stock of debt B_t. Each period, the government pays the nominal interest rate r on the outstanding stock of debt. Total government financial outflows equals payments on the debt, rB_t, plus government outlays, G_t, which consists of both government spending and government transfers. Government inflows come from tax revenue T_t, issuing new bonds $\Delta B_{t+1} = B_{t+1} - B_t$, and creating money $\Delta M_{t+1} = M_{t+1} - M_t$. Since inflows equal outflows in each period, the budget constraint is

$$G_t + rB_t = T_t + \Delta B_{t+1} + \Delta M_{t+1}.$$

As in Jones (2024), I assume that the government cannot print money to finance spending, so $\Delta M_{t+1} = 0$. Rearranging, we find the key equation in Jones (2024),

$$\Delta B_{t+1} = \underbrace{rB_t + \underbrace{G_t - T_t}_{\text{Primary deficit}}}_{\text{Total deficit}}. \tag{1}$$

The primary deficit is defined as $G_t - T_t$; the total deficit, which includes interest payments, is $rB_t + G_t - T_t$. From Equation (1), students can observe that a positive total deficit necessarily implies an increase in the stock of government debt. In contrast, to reduce debt, the government must run a total surplus.

At this point, I depart from Jones (2024). While Equation (1) describes the law of motion for the debt level, B_t, my theory is based on the law of motion for the debt-to-nominal GDP *ratio*, B_t/Y_t. First, isolate B_{t+1} on the left hand side,

$$B_{t+1} = B_t(1+r) + G_t - T_t,$$

and divide both sides by Y_t to obtain

$$\frac{B_{t+1}}{Y_t} = \frac{B_t}{Y_t}(1+r) + \frac{G_t - T_t}{Y_t}. \tag{2}$$

From here, I impose two additional assumptions: (1) GDP grows at a constant rate g, so $Y_{t+1} = (1+g)Y_t$, and (2) the primary deficit-to-GDP ratio is constant at $d = \frac{G_t - T_t}{Y_t}$.[1] Plug in $Y_t = Y_{t+1}/(1+g)$ on the left side and $d = \frac{G_t - T_t}{Y_t}$ on the right side of Equation (2) to obtain

$$\frac{B_{t+1}}{Y_{t+1}}(1+g) = \frac{B_t}{Y_t}(1+r) + d.$$

I denote the debt-to-GDP ratio as $b_t = B_t/Y_t$. Using b_t and dividing both sides by $1+g$, we arrive at

$$b_{t+1} = b_t\left(\frac{1+r}{1+g}\right) + \frac{d}{1+g}.$$

It is convenient to isolate the *change* in the debt-to-GDP ratio, $\Delta b_{t+1}(b_t) = b_{t+1} - b_t$, on the left side. One simple way to get there is to subtract b_t from both sides:

$$\Delta b_{t+1}(b_t) = b_{t+1} - b_t = b_t\left(\frac{1+r}{1+g} - 1\right) + \frac{d}{1+g}.$$

Finally, rewriting the term inside the parentheses, we arrive at our main dynamic equation:

$$\Delta b_{t+1}(b_t) = b_t\left(\frac{r-g}{1+g}\right) + \frac{d}{1+g}. \tag{3}$$

Given that the interest rate is constant for now, Equation (3) is the "constant r" version of the model.[2] Finally, assume that the debt-to-GDP ratio begins at b_0 in time $t = 0$.

[1] Equivalently, we could say that G_t and T_t also grow at the rate g.

[2] Like Carlin and Soskice (2023), one could use $\frac{r-g}{1+g} \approx r - g$, though students may find the approximation confusing.

2.2 Government Budget Sustainability under Constant Interest Rates

What precisely does it mean if the government budget is unsustainable? Though typically an overcomplication in intermediate macroeconomics, it is clarifying to define sustainability mathematically. Under constant interest rates, the government budget consists of four variables: (1) the initial debt-to-GDP ratio, b_0, (2) the primary deficit-to-GDP ratio, d, (3) the growth rate of nominal GDP, g, and (4) the nominal interest rate, r. In this paper, the government budget is considered unsustainable if the debt-to-GDP ratio is predicted to increase forever until it "explodes" to infinity. Otherwise, the government budget is sustainable. Mathematically, I define unsustainability as in Definition 1:

Definition 1 *Given a constant interest rate, the government budget* $\{b_0, d, g, r\}$ *is unsustainable if* $b_t \to \infty$ *as* $t \to \infty$.

Students may be surprised that sustainability does not require the elimination of debt. One can intuitively explain that many firms and households are in debt, but they are not considered to be in serious financial trouble if the debt is expected to be paid off without difficulty. Their situation only becomes unsustainable if debt begins to spiral out of control.[3] The same is true for the government. If the debt-to-GDP ratio perpetually increases, the government will never be able to pay off its debt and must eventually default.

Under this definition, sustainability comes down to a simple condition: any government budget is sustainable as long as $r < g$.[4] To understand why, think of Equation (3) as a linear equation in the form of $y = mx + b$: the y variable is Δb_{t+1}, the x variable is b_t, the slope is $\frac{r-g}{1+g}$, and the y intercept is is $\frac{d}{1+g}$. I plot this line twice in Figure 1. I draw each line with the same y-intercept, so the only difference across panels is the slope. In Panel A, $r < g$, which implies that the slope, $\frac{r-g}{1+g}$, is negative. In Panel B, $r > g$, so the slope is positive.

[3] Governments have the added advantage of living infinitely. People eventually die; governments have no such constraint.

[4] I ignore the cases where the deficit is zero or negative since they are less relevant for most countries.

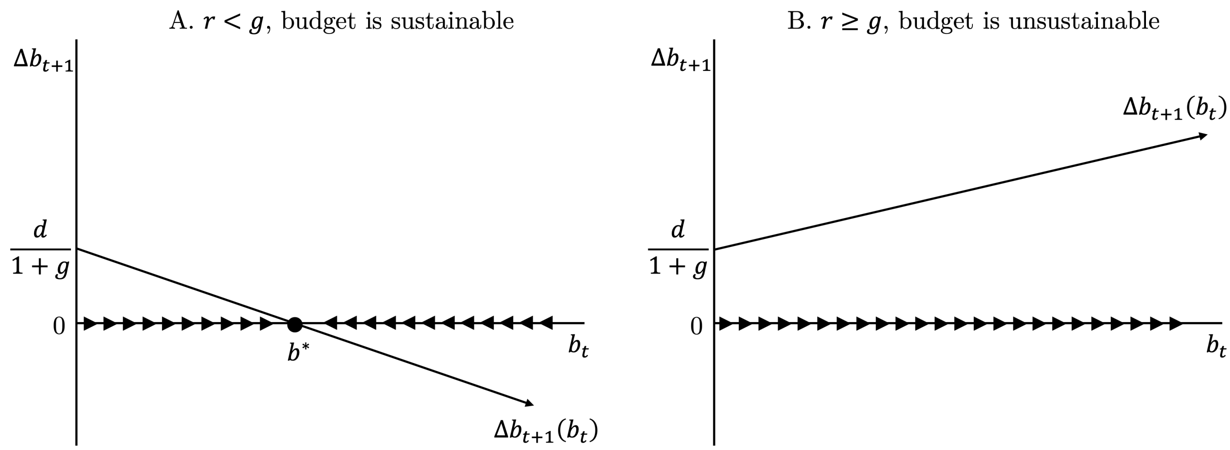

Figure 1: Evolution of debt-to-GDP ratio with constant interest rate

Plots follow Equation (3). Each panel is drawn with the same y-intercept. In Panel A, $r < g$, so the slope of the line is negative. Therefore, the debt-to-GDP ratio will converge to a steady state, b^*, and the budget is sustainable. In Panel B, $r > g$, so the slope of the line is positive. Therefore, the debt-to-GDP ratio will increase until it becomes infinitely large, and the budget is unsustainable.

Consider the case when $r < g$ in Panel A. Since the slope is negative and the y-intercept is positive, the $\Delta b_{t+1}(b_t)$ line is guaranteed to intersect the x-axis at some point. Furthermore, the model predicts that the debt-to-GDP ratio will eventually converge to this point. I denote the intersection as b^* and refer to it as the steady-state debt-to-GDP ratio.

Why does the debt-to-GDP ratio converge to a steady state? The logic is the same as in the standard Solow diagram. First, suppose that b_t begins at some level lower than b^*. At this level of b_t, the $\Delta b_{t+1}(b_t)$ curve is above the x-axis, so $\Delta b_{t+1}(b_t) > 0$, and the debt-to-GDP ratio will increase. Alternatively, suppose instead that b_t begins at a level above b^*. In that case, $\Delta b_{t+1}(b_t)$ is negative, and the debt-to-GDP ratio will decrease. Finally, if $b_t = b^*$, then $\Delta b_{t+1}(b_t) = 0$, and the debt-to-GDP ratio will remain constant. Therefore, regardless of where the economy begins, if $r < g$, the debt-to-GDP ratio will converge to a steady state. In fact, whether government debt is sustainable or not does not depend on the size of the budget deficit d or the initial debt b_0.

It is crucial that students understand the intuition. At the steady state, since Y_t grows at the rate g, then B_t must also grow at the rate g. So, the stock of debt is perpetually increasing. How is that sustainable? Since tax revenue is a constant fraction of GDP, the government earns more tax revenue as the economy grows. In the steady state, the increase in tax revenue is just large enough to pay off enough debt so that B_t does not grow faster than Y_t. The debt that is not paid off is rolled over; that is, the government pays off old bonds by issuing new bonds. Put simply, the current generation can borrow against their children's future, who will borrow against their children's future, and so on. It is neither procrastinating default nor a Ponzi scheme; since lenders are always paid on time, they will be happy to continue loaning to the government.

Finally, students can easily solve for b^* analytically. To do so, solve for b^* such that $\Delta b_{t+1}(b^*) = 0$, which yields

$$b^* = \frac{d}{g - r}. \tag{4}$$

Equation (4) is intuitive. To run a larger primary deficit, the government must roll over more debt every period, so b^* must increase. If the economy grows faster, then the government does not need to roll over as much debt every period; the opposite is true if interest rates are high.

Contrast Panel A with Panel B, which illustrates the case where $r > g$ (the same results also hold for $r = g$). Since the slope of $\Delta b_{t+1}(b_t)$, $\frac{r-g}{1+g}$, is positive, the $\Delta b_{t+1}(b_t)$ line never crosses the x-axis. Thus, no matter how much debt the government starts with, b_t will increase until it becomes infinitely large and impossible to pay. Thus, if $r \geq g$, the government cannot sustain *any* positive primary deficit, and must run a primary surplus or balanced budget.

To summarize, with constant interest rates, any government budget is sustainable as long as $r < g$. Students will recognize that this is a peculiar conclusion. After all, it implies unlimited free lunch; in theory, the government could potentially fund the entire economy ($G_t/Y_t = 1$), never impose taxes ($T_t/Y_t = 0$, so $d = 1$), and do it all with debt, rolling over the debt into perpetuity.

2.3 Government Budget Sustainability Under Endogenous Interest Rates

Above, I assumed that the government can borrow any sum it wants without consequences. To make the analysis more realistic, we must acknowledge that, in some sense, as the government borrows more, it will face increasing difficulty in borrowing additional funds. This concept can be modeled in several ways. Here, I incorporate the empirical regularity that the interest rate rises endogenously with the debt-to-GDP ratio, $r(b_t)$.[5]

There are several reasons to expect that the interest rate on government debt increases with the size of government debt. For the purposes of this extension, I prefer the intuitive risk-premium story: as the government increases its debt, lenders become increasingly concerned that the government will not be able to repay them, so they demand higher interest rates to compensate for the additional risk.[6]

For simplicity, I assume that the interest rate is a linear function of government debt. Given the current nominal interest rate r_0 and debt-to-GDP ratio b_0,

$$r(b_t) = \beta(b_t - b_0) + r_0 \tag{5}$$

where β represents the sensitivity of the interest rate to the debt level. Equation (5) implies that if the debt-to-GDP ratio increases (or decreases) by 1 percentage point, then the interest rate increases (or decreases) by β percentage points.

Returning to the dynamics of the debt-to-GDP ratio, we rewrite Equation (3) as

$$\Delta b_{t+1}(b_t) = b_t \left(\frac{r(b_t) - g}{1+g} \right) + \frac{d}{1+g} = \frac{\beta b_t^2 + b_t(r_0 - g - \beta b_0)}{1+g} + \frac{d}{1+g} \tag{6}$$

[5] My analysis departs from Carlin and Soskice (2023) at this point.

[6] Another explanation is crowding out: an increase in government borrowing soaks up loanable funds, increasing the equilibrium interest rate. Convenience utility offers a more complex explanation: the idea is that government debt has special convenience properties, but these properties exhibit decreasing returns, so lenders require higher interest rates to hold more. Finally, though it is assumed away in the model, the government retains the ability to "inflate away the debt" by printing money. If lenders expect more inflation in the future, the nominal interest rate must increase for the real interest rate to remain fixed.

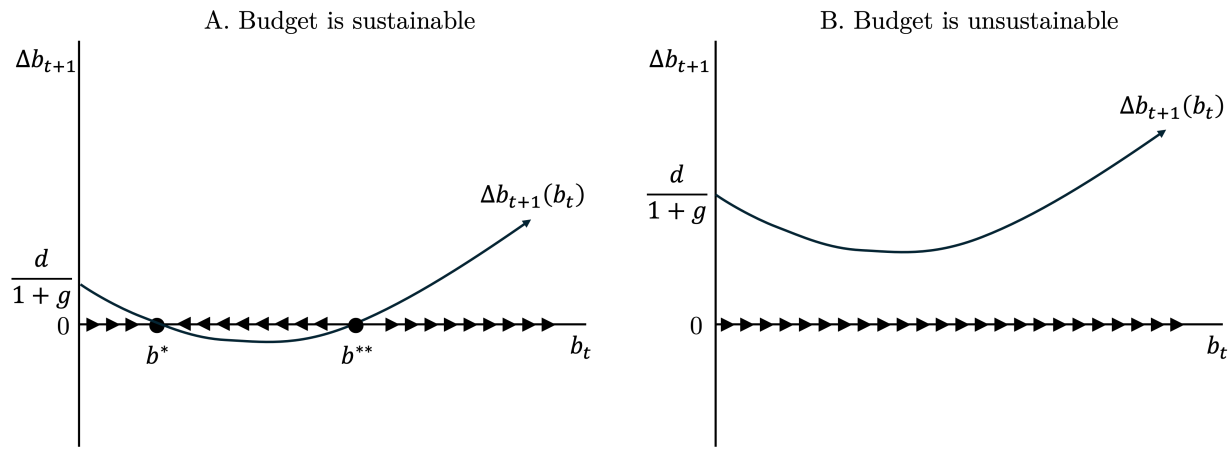

Figure 2: Evolution of debt-to-GDP ratio with endogenous interest rate

Plots follow Equation (6). In Panel A, the $\Delta b_{t+1}(b_t)$ curve crosses the x-axis at b^* and $b^{*}*$. Both points are steady states. If the debt-to-GDP ratio begins between 0 and $b^{*}*$, it will eventually converge to b^*, which implies that the budget is sustainable. If it starts above $b^{*}*$, it will increase indefinitely, and the budget is unsustainable. In Panel B, the $\Delta b_{t+1}(b_t)$ lies above the x-axis. Therefore, the debt-to-GDP ratio will increase indefinitely, and the budget is unsustainable.

where $r(b_t)$ is given by Equation (5).

Now, the government budget consists of five variables: (1) the initial debt-to-GDP ratio, b_0, (2) the primary deficit-to-GDP ratio, d, (3) the growth rate of nominal GDP, g, (4) the initial nominal interest rate, r_0, and (5) the sensitivity of the interest rate to the debt-to-GDP ratio, β. As before, we define unsustainability as in Definition 2:

Definition 2 *Given an endogenous interest rate, the government budget* $\{d, r_0, b_0, g, \beta\}$ *is unsustainable if* $b_t \to \infty$ *as* $t \to \infty$*.*

Endogenous interest rates significantly alter the analysis. The new $\Delta b_{t+1}(b_t)$ functions are drawn twice in Figure 2. First, note that $\Delta b_{t+1}(b_t)$ is nonlinear. Mathematically, we know that if b_t is large enough, the slope of b_t will eventually become increasingly positive. Beyond that fact, I draw the $\Delta b_{t+1}(b_t)$ curves in Figure 2 deliberately. First, I assume that if there is no debt, then $r(0) < g$, so the curve slopes downward from the y-axis. Thus, there are low values of b_t where $r(b_t) < g$. But, as b_t increases, the interest rate increases with it, so the $\Delta b_{t+1}(b_t)$ curve eventually slopes upward. Second, the curves in Panels A and B are identical except that the primary deficit, d, is greater in Panel B, which moves the curve upward. Again, I have drawn two scenarios, one with a sustainable government budget and one without.

Panel A illustrates a potentially sustainable budget. In Panel A, the $\Delta b_{t+1}(b_t)$ curve crosses the x-axis twice, at b^* and b^{**}. If the debt-to-GDP ratio begins below b^*, then since $\Delta b_{t+1}(b_t) > 0$, b_t will increase; if the debt-to-GDP ratio begins between b^* and b^{**}, then since $\Delta b_{t+1}(b_t) < 0$, it will decrease. Thus, in the region from 0 up to b^{**}, the debt-to-GDP ratio will converge to b^*. So, b^* is a steady state, and if b_0 begins between 0 and b^{**}, the government budget is sustainable. Since $\Delta b_{t+1}(b^{**}) = 0$, b^{**} is also a steady state, but the economy only converges to b^{**} if it begins at b^{**}. Thus, with endogenous interest rates, b_0 also matters for sustainability. If the debt-to-GDP ratio starts too high, $b_0 > b^{**}$, then it is predicted to increase until it spirals out of control.

Panel B illustrates a scenario where the government budget is unsustainable. Because d is large, $\Delta b_{t+1}(b_t)$ is always greater than zero, and the debt-to-GDP ratio only increases.

To summarize, if the interest rate on government debt is endogenous, then $r_0 < g$ today does not imply that the government budget will be sustainable in the future. Furthermore, sustainability no longer depends only on r and g, but also on the primary deficit-to-GDP ratio, d, and the current debt-to-GDP ratio, b_0. Nevertheless, as long as the government budget meets certain conditions, it can still sustain a positive primary deficit.

2.4 Maximum Sustainable Primary Deficit

In the quantitative section below, I show that the current US government budget is unsustainable. Students will naturally wonder what changes to taxes and spending would put the government on a sustainable path. In my framework, we can use simple calculus to analytically solve for the maximum primary deficit-to-GDP ratio that would make the budget sustainable, d_{max}. The government must then increase taxes or decrease spending to bring the primary deficit down to d_{max}.[7]

[7] In practice, it may be wise to teach this part after going through some of the quantitative analysis below to break up the theory and data.

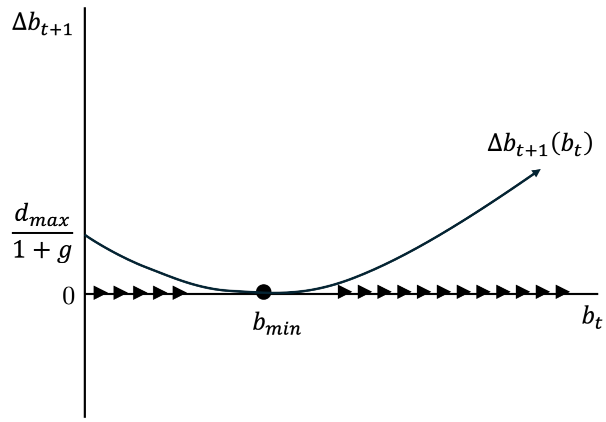

Figure 3: Evolution of debt-to-GDP ratio with endogenous interest rate

Given the rest of the government budget, d_{max} is the maximum sustainable primary deficit. To calculate d_{max}, find b_{min}, the minimum of $\Delta b_{t+1}(b_t)$, then solve for d such that $\Delta b_{t+1}(b_{min}) = 0$.

Refer to the U-shaped curves in Figure 2. We can see that the government budget is sustainable only if $\Delta b_{t+1}(b_t)$ touches the x-axis at some point.[8] Mathematically, the deficit is sustainable if and only if there is a value of b_t such that $\Delta b_{t+1}(b_t) \leq 0$. This condition is guaranteed if the minimum of the $\Delta b_{t+1}(b_t)$ curve is less than or equal to zero. See Figure 3.

Using calculus, we can find the minimum of the $\Delta b_{t+1}(b_t)$ curve. Using Equation (6), we first solve $\frac{d\Delta b_{t+1}(b_t)}{db_t} = 0$, which results in

$$\frac{d\Delta b_{t+1}(b_t)}{db_t} = \frac{2\beta b_t + r_0 - g - \beta b_0}{1+g} = 0.$$

Let b_{min} denote the value of b_t that minimizes $\Delta b_{t+1}(b_t)$. After some algebra, we find

$$b_{min} = \frac{\beta b_0 - r_0 + g}{2\beta}. \tag{7}$$

The budget is sustainable if $\Delta b_{t+1}(b_{min}) \leq 0$. To find $\Delta b_{t+1}(b_{min})$, plug Equation (7) into Equation (6). After a little more algebra, we find that the government budget is sustainable

[8] Assume that the initial level of debt b_0 starts sufficiently low. Interestingly, Japan violates this assumption.

if

$$d \leq \frac{(g + \beta b_0 - r_0)^2}{4\beta}. \tag{8}$$

In other words, the maximum possible deficit that the government can sustain is

$$d_{max} = \frac{(g + \beta b_0 - r_0)^2}{4\beta}. \tag{9}$$

If $d = d_{max}$, then $\Delta b_{t+1}(b_t)$ is tangent to the x-axis. If $d < d_{max}$, then $\Delta b_{t+1}(b_t)$ crosses the x-axis twice, as in Panel A of Figure 2.

Note that d_{max} is increasing in g and decreasing in r_0. Thus, if the government cannot decrease d, it can make its budget more sustainable by increasing growth or decreasing the interest rate.

3 Quantitative Analysis

3.1 Current Situation of the US Government

Before taking the model to the data, students will benefit from an overview of the US government budget such as Figure 4. In the notes below each graph in this section, I provide links to up-to-date versions of those graphs in FRED. The data in Figure 4 reflects only the federal government, but I prefer it because it is internally consistent and goes further back in time. See the footnotes for links to data that aggregates across federal, state, and local governments.[9]

Panel A plots the debt-to-GDP ratio, b_t, over time.[10] The debt-to-GDP ratio has been rapidly increasing since 2000, spurred on by the Great Financial Crisis and COVID-19. As of 2025, the federal debt-to-GDP ratio is approaching 100%, a level not seen since World

[9] The same graphs for combined federal, state, and local governments are available at https://fred.stlouisfed.org/graph/?g=1X49B, https://fred.stlouisfed.org/graph/?g=1X49N, and https://fred.stlouisfed.org/graph/?g=1X49T. Source for flows and GDP: BEA, NIPA Tables 1.1.5 and 3.1. Source for debt levels: Federal Reserve Board of Governors, Financial Accounts of the US (Z.1).

[10] Government debt is calculated as debt held by the public. See FRED Blog (2019) for an explanation of who holds federal government debt.

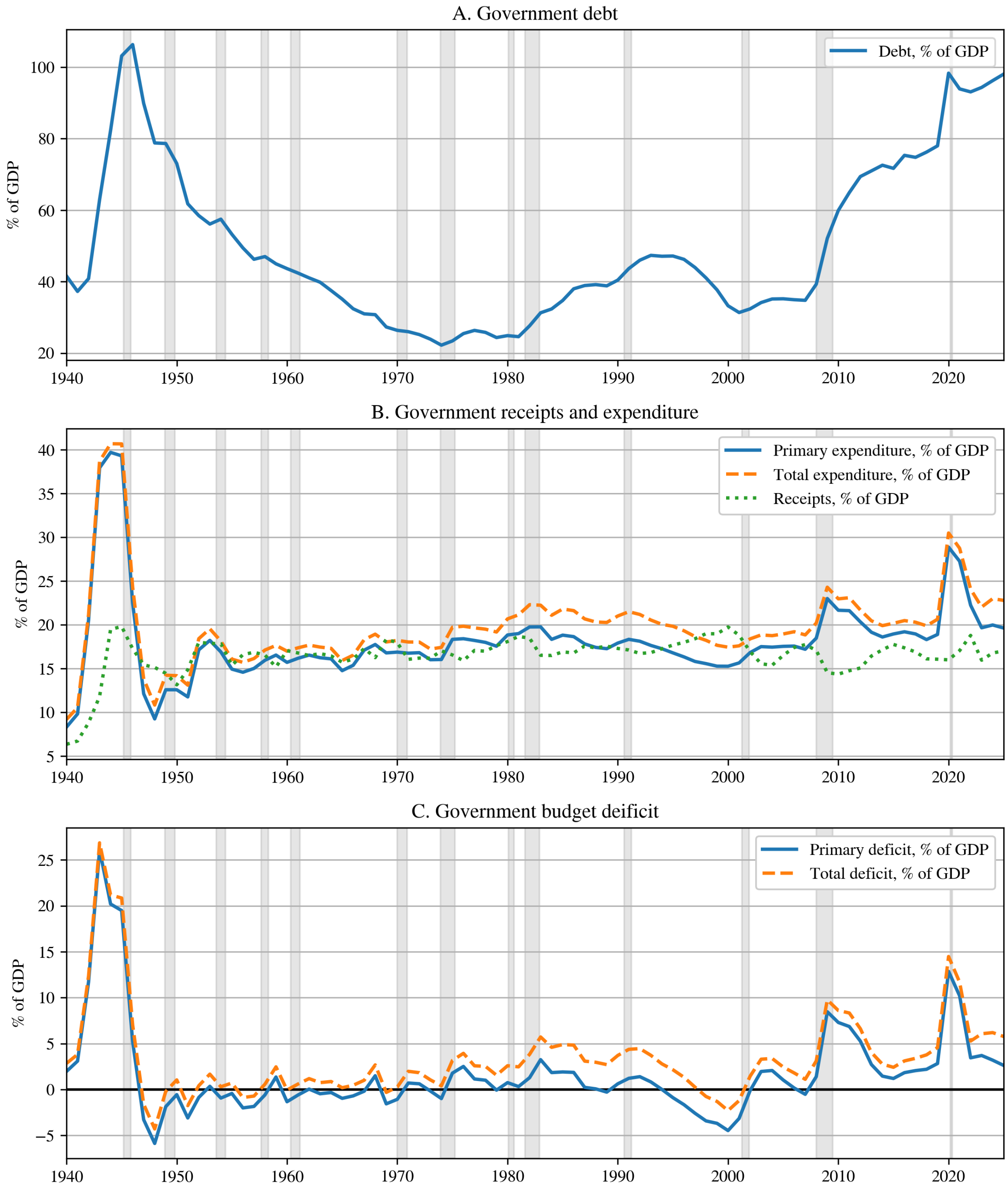

Figure 4: Federal US government budget over time

Source: US Office of Management and Budget. All data is annual. Debt is defined as total government debt held by the public. Up-to-date figures are available from FRED at https://fred.stlouisfed.org/series/FYPUGDA188S, https://fred.stlouisfed.org/graph/?g=1X45F, and https://fred.stlouisfed.org/graph/?g=1X46D.

War II, and by far the largest in peace time in the history of the US.[11]

As shown by Equation (3), the amount of government debt is determined by government inflows and outflows. These flows are plotted in Panel B as primary expenditures, total expenditures, and receipts as percentages of GDP. (Recall that primary and total expenditures are equivalent except that primary expenditures exclude interest payments.) By far, expenditures were largest in World War II. Since 1950, expenditures and tax revenue have fluctuated between 15% and 20% of GDP. Since 2000, government expenditures have increased, again spurred on by the Great Financial Crisis and COVID-19, while tax revenue has slightly decreased.

Primary and total deficits are plotted in Panel C. The gap between primary expenditures and receipts equals the primary deficit, and the gap between total expenditures and receipts equals the total deficit. Between 1950 and 1970, both deficits were close to zero. Both deficits then increased in the 1970s and 1980s before the federal government temporarily ran a budget surplus in 2000. Since then, increased spending and slightly decreased revenue have increased both deficits. Currently, the gap between the primary deficit and the total deficit is widening, reflecting that a larger share of government outflows is going toward interest payments.

In this paper, students determine whether current deficits are sustainable. However, for further context, it may be worth considering how these deficits are expected to change in the future. In 2025, the total deficit was 5.8%, and the primary deficit was 2.6% of GDP. The Congressional Budget Office (CBO) forecasts that by 2036, the primary deficit will slightly decline to 2.1% of GDP while, due to higher interest rates, the total deficit will continue to increase to 6.7% of GDP (Congressional Budget Office, 2026).[12] As such, the CBO projects

[11] Data on government debt dating back to the US's founding in Mankiw (2025) confirm that the current level of debt is the second-highest in the country's history, second only to World War II. Before World War II, the debt-to-GDP ratio only reached about 40% in the Revolutionary War, Civil War, World War I, and the Great Depression.

[12] The CBO predicts that the primary deficit will slightly decrease because tax revenue as a percentage of GDP will slightly increase while primary expenditure remains about the same. CBO also predicts that primary expenditures will remain roughly flat for the next decade because increased mandatory spending on Social Security and Medicare will roughly cancel out with decreased discretionary spending.

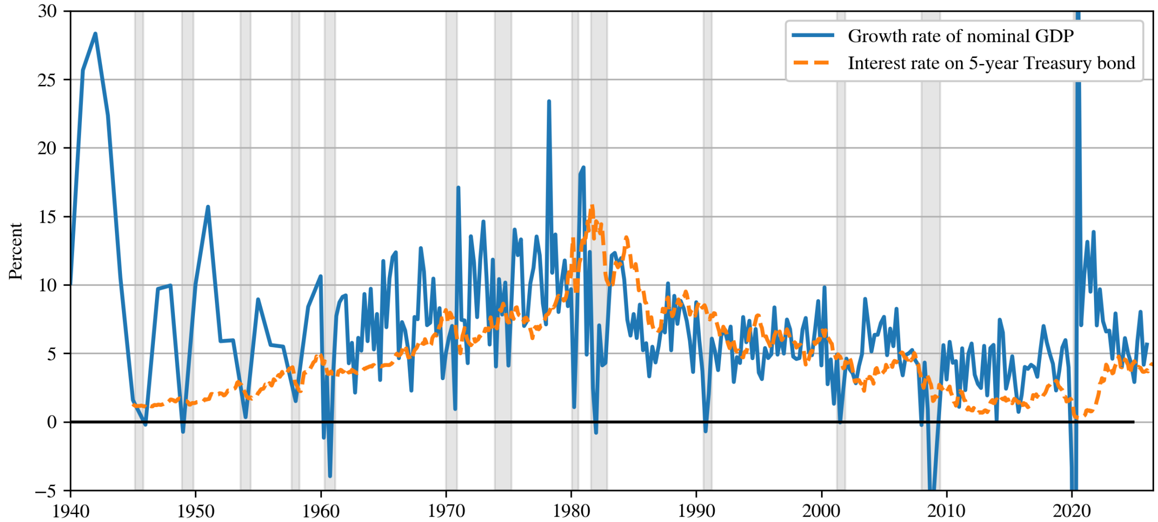

Figure 5: r and g in US data

GDP source: Bureau of Economic Analysis. Interest rate source prior to 1962: NBER Macrohistory Database. Interest rate source since 1962: Federal Reserve Board of Governors H.15 Statistical Release. GDP data is annual prior to 1960 and quarterly thereafter; interest rate data is monthly. Up-to-date data for this graph is available at https://fred.stlouisfed.org/graph/?g=1X4h3.

that federal government debt will increase to 120% of GDP by 2036 and 175% by 2056.[13]

Budget sustainability depends not only on debt and deficits, but also on the interest rate on government debt and the growth rate of nominal GDP, which are plotted in Figure 5. Generally, $r < g$ usually holds, but it is not guaranteed. In fact, at the moment, there is little difference r and g. As stated in Ball et al. (1998), relying on $r < g$ is a "gamble." Simultaneously, given the very low interest rates, it is understandable that the $r < g$ debate intensified during the 2010s. Though the Fed's response to the Great Financial Crisis is a significant reason for the low interest rates in the 2010s, it is also clear that interest rates were trending downward prior.

Table 1: US benchmark parameters

Parameter	Meaning	Value	Notes
b_0	Current debt-to-GDP ratio	98.1%	Federal only, fiscal year 2025
d	Primary deficit-to-GDP ratio	2.6%	Federal only, fiscal year 2025
g	Growth rate of nominal GDP	5.5%	Mean for 2016-2025
r_0	Current nominal interest rate	4.28%	Mean yield on 5-year Treasury note for July 2026
β	Sensitivity of r to debt-to-GDP ratio	0.025	See Table 2

Parameters used in benchmark analysis of US government debt. Source for b_0 and d: US Office of Management and Budget. Source for g: Bureau of Economic Analysis. Source for r_0: Federal Reserve Board of Governors.

3.2 Parameterization

At this point, students have the tools to quantitatively analyze the sustainability of government budgets. What remains is to pick parameter values. For my benchmark US results, I use the parameter values in Table 1. Since this paper seeks to answer whether the current budget is sustainable, I use the most recent federal figures for the primary deficit-to-GDP ratio, debt-to-GDP ratio, and interest rate on government debt. For the interest rate, I use the interest rate on the 5-year Treasury note, which is roughly the weighted average maturity of outstanding US Treasury bonds (Anderson, 2026).[14]

As Figure 5 illustrates, the growth rate of nominal GDP is volatile, so the most recent value of GDP growth may provide limited guidance for forecasting future growth. I opt to take the mean growth rate over the last ten years. This stretch covers the slow, steady growth of 2016-2020 and the inflation-fueled post-COVID growth.

The only parameter not easily available in the data is β, the sensitivity of the interest rate to the debt-to-GDP ratio. Estimating the effect of government debt on interest rates is a difficult task, and students (especially those who have taken econometrics) will recognize the challenge. Fortunately, many papers try to estimate β. Table 2 lists the headline estimated β values of several recent papers. I use a representative value $\beta = 0.025$, which implies

[13] This paragraph is based on Figures 1-1, 3-1, and 1-8 in Congressional Budget Office (2026).

[14] I measure interest rates using the market yield to constant maturity on an investment basis. Given that new bonds are not issued every day, it may be useful to explain that this measures the yield that a theoretical, brand-new bond would yield would offer if it were issued today.

Table 2: β values in the literature

Paper	β
Plante et al. (2025)	0.030
Chadha et al. (2025)	0.017
Neveu and Schafer (2024)	0.020
Gamber and Seliski (2019)	0.019
Laubach (2009)	0.033
Engen and Hubbard (2005)	0.030

Estimates for β, the sensitivity of the interest rate on government debt to the debt-to-GDP ratio, from the literature. I use $\beta = 0.025$.

that if the debt-to-GDP ratio increases by one percentage point, then the interest rate on government debt will increase by 0.025 percentage points (2.5 basis points).[15]

3.3 US Benchmark Results

In Figure 6, I compute quantitative versions of the dynamic debt-to-GDP diagrams using the US benchmark parameters in Table 1. Panels A and B illustrate the constant-r and endogenous-r versions of the model. In both cases, I plot the 2025 value of b_t and its corresponding $\Delta b_{t+1}(b_t)$.

If we assume the interest rate is constant, as in Panel A, then because $r < g$, the current government budget is sustainable. The model predicts that the debt-to-GDP ratio will increase and eventually converge to about 200%. Although certainly large, a debt-to-GDP ratio of 200% is not unheard of; Japan has maintained that amount of debt since the 2010s.

However, the outlook is less optimistic when I incorporate endogenous interest rates in Panel B. Now, the theory predicts that as the debt-to-GDP ratio increases, the interest rate will increase so fast that $\Delta b_{t+1}(b_t)$ never crosses the x-axis, implying that the debt burden will eventually spiral out of control.

Using Equation (9) and the parameter values in Table 1, I find that the primary deficit-

[15]See also the table in Gust and Skaperdas (2024). For an international perspective, see Gruber and Kamin (2012), Kumar and Baldacci (2010), Ardagna et al. (2007), and Kinoshita (2006). Mian et al. (2025) use a clever method to estimate β using the 2021 Georgia Senate run-off election.

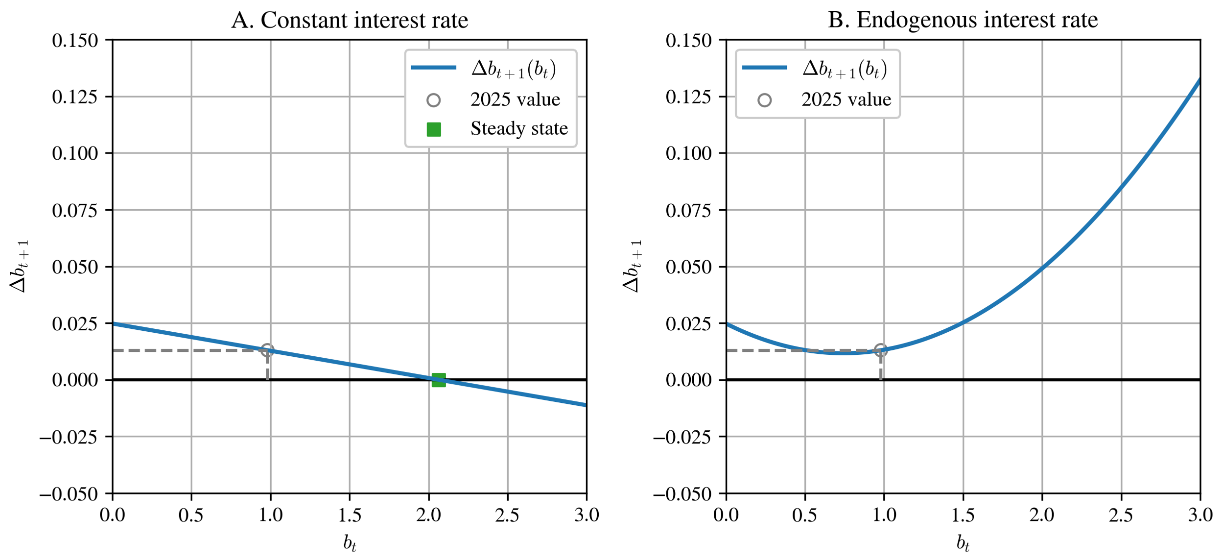

Figure 6: US benchmark results for 2025

Quantitative computations of Figures 1 and 2 using US parameter values in Table 1. Panel A plots the model with constant interest rates given by Equation (3). Since $r < g$ in the data, the $\Delta b_{t+1}(b_t)$ line slopes downward, and the model predicts that the debt-to-GDP ratio will eventually converge to the steady state. Therefore, this version of the model suggests that the current budget is sustainable. Panel B plots the model with endogenous interest rates given by Equation (6). Even though $r < g$ in the data, this version of the model predicts that the interest rate will increase fast enough that the debt-to-GDP ratio will eventually diverges to infinity. Therefore, this version of the model suggests that the current budget is unsustainable.

to-GDP ratio that would make the budget sustainable is approximately 1.4%, almost half of the current primary deficit. Reaching such a low deficit would require considerable sacrifices in the form of higher taxes or less government spending.

However, the US can also achieve sustainability by increasing economic growth or decreasing interest rates. In Figure 7, I compute the sensitivity of d_{max} by varying one parameter at a time. The vertical lines are the benchmark parameter values, and the horizontal lines show the current value of d. Therefore, the current primary deficit could be sustainable if the growth rate of nominal GDP is 7% instead of 5.5%, or if the interest rate on government debt is 3% instead of 4.3%.

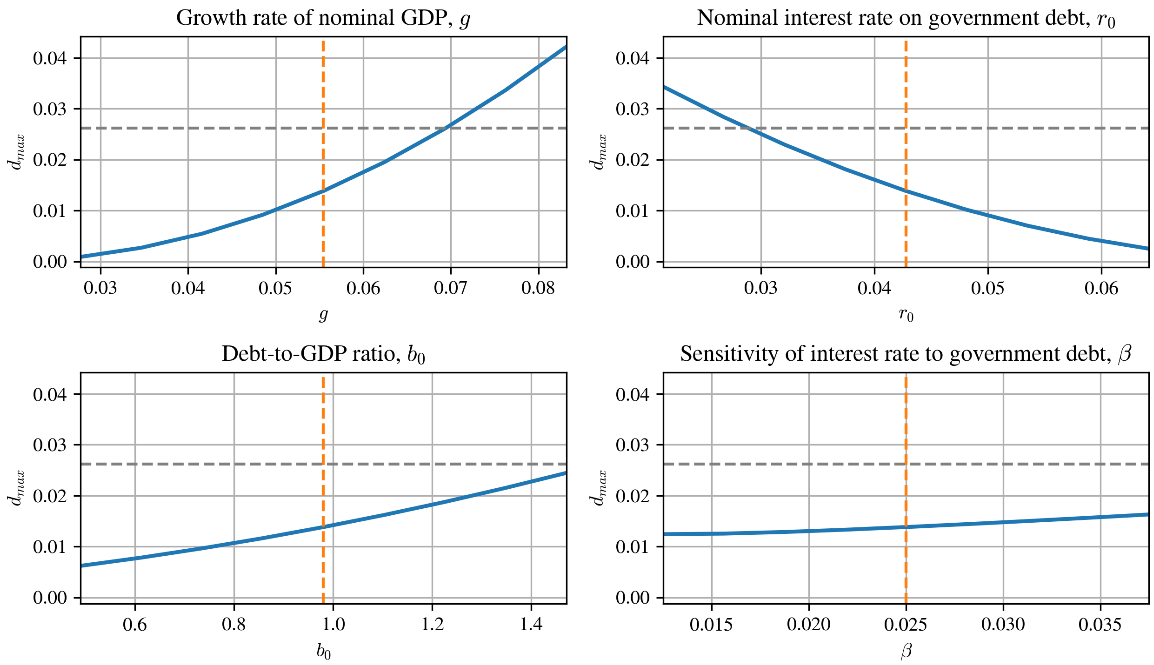

Figure 7: Sensitivity of d_{max} to parameters in US benchmark

Computes d_{max}, the maximum sustainable primary deficit, for the US using Equation (9) while varying one parameter at a time. Each parameter is taken to 50% below and 50% above its benchmark value. The vertical dashed lines are the current parameter values. The horizontal dashed lines show the current value of d.

Table 3: International results

Country	r_0	g	b_0	T/Y	G/Y	d	d_{max}	Sustainable
	2026							
Australia	4.99	5.73	51.03	36.67	38.39	1.72	0.40	No
Brazil	9.13	7.91	93.33	39.35	39.74	0.39	0.12	No
Canada	3.54	5.12	113.51	42.19	44.28	2.09	1.95	No
China	1.75	7.17	99.24	25.01	31.92	6.91	6.25	No
France	3.74	3.14	115.99	52.35	55.48	3.13	0.53	No
Germany	3.05	3.81	62.93	47.88	49.66	1.78	0.54	No
Italy	3.82	3.21	137.09	47.94	47.45	-0.50	0.80	Yes
Japan	2.65	1.94	206.53	35.80	36.66	0.85	1.99	Yes
Mexico	9.45	6.33	61.75	24.98	23.75	-1.23	0.25	Yes
South Korea	4.08	4.36	52.25	22.56	24.04	1.49	0.25	No
United Kingdom	4.94	4.74	102.32	38.27	41.03	2.77	0.55	No
United States	4.48	5.38	123.89	30.88	34.05	3.17	1.60	No

Country	r_0	g	b_0	T/Y	G/Y	d	d_{max}	Sustainable
	2016							
Australia	1.94	5.52	37.58	34.34	36.20	1.86	2.04	Yes
Brazil	7.50	10.75	71.73	36.91	37.76	0.85	2.54	Yes
Canada	1.05	3.48	92.28	39.96	39.37	-0.59	2.24	Yes
China	2.78	14.08	40.78	28.51	30.51	2.00	15.17	Yes
France	0.17	2.27	96.98	53.71	55.66	1.95	2.05	Yes
Germany	-0.15	2.90	71.18	45.40	43.41	-2.00	2.34	Yes
Italy	1.23	1.07	134.79	47.78	46.31	-1.48	1.03	Yes
Japan	-0.20	0.25	200.15	32.94	35.56	2.62	2.97	Yes
Mexico	6.08	6.82	50.97	22.70	23.86	1.17	0.40	No
South Korea	1.40	5.77	38.84	19.30	19.12	-0.18	2.85	Yes
United Kingdom	0.96	3.25	87.34	35.55	38.58	3.03	2.00	No
United States	1.50	3.47	105.44	31.50	33.19	1.69	2.12	Yes

Parameter values and model predictions for a sample of countries for 2026 and 2016. All numbers are expressed as percentages. Moving left to right, columns refer to the current nominal interest rate on government debt, the mean growth rate of nominal GDP for the previous ten years, the current debt-to-GDP ratio, tax revenue as a fraction of nominal GDP, government expenditure as a function of GDP, the primary deficit-to-GDP ratio, the model-implied maximum sustainable primary deficit-to-GDP ratio, and whether the current budget is sustainable according to the model. The interest rate is for the country's ten-year government bonds. Interest rate source: OECD Financial Indicators. Source for rest of the data: IMF World Economic Outlook.

3.4 International Results

We can extend the analysis across countries and time. Table 3 lists the parameter values for a sample of countries at two points in time, current (2026) and ten years ago (2016).[16] As before, the growth rate of nominal GDP is averaged across the ten years preceding. I also include data on tax revenue and government expenditures as a percent of nominal GDP.[17] Then, for each country, I use the endogenous-r version of the model to calculate the maximum primary deficit, which determines if the current primary deficit is sustainable. Most of the countries in Table 1 are rich countries, which tend to have similar β values; in general, less developed countries will have higher β.

There are several points of interest in Table 3. First, my model suggests that the US government was sustainable in 2016. In fact, my whole sample of countries has tended toward unsustainability since then. What changed? In the US, there have been two major developments. Most importantly, the primary deficit-to-GDP ratio has roughly doubled. Second, both r and g have increased due to recent inflation, and although $r < g$ still holds, the gap between r and g has narrowed, leaving the government with less room to maneuver.

Other notable examples include China, which shows the importance of economic growth. If Chinese growth had remained as strong before 2026 as it was before 2016, it could sustain its massive current deficit. Japan in 2016 is another interesting case. Although Japan had a large 200% debt-to-GDP ratio and low growth, due to negative interest rates, the interest rate was even lower than the growth rate, and Japan could theoretically sustain a modest deficit.

[16] Each date is comfortably between recessions.

[17] For ease of comparison, I adjust my data collection in two ways. First, I use data on "general government" finances, which combines all levels of government. In the US, that refers to the federal, state, and local levels. It also includes some cross-agency debt. Thus, debt levels in the US appear higher in this data. Second, I use the interest rate on the global benchmark 10-year government bonds. Interest rate data comes from the OECD, and the rest of the data comes from the IMF World Economic Outlook (WEO).

4 Conclusion

In this paper, I develop a simple framework for introducing students in Intermediate Macroeconomics to the $r < g$ debate. Using the intertemporal government budget constraint, steady-state reasoning, and diagrams, students learn why $r < g$ suggests that the government can borrow infinitely. Introducing a mechanism where the interest rate increases with government debt sharpens the analysis and shows why $r < g$ does not imply an unlimited free lunch. According to the model, most rich countries, including the US, face an unsustainable fiscal path.

Appendix

A Sample Questions

1. All of the theory has been in nominal terms. However, we know that the government is tempted to "inflate away" debt by printing money. In this model, does increasing inflation make the budget more sustainable? To answer, decompose the growth rate of nominal GDP into the growth rate of real GDP, G, and inflation, π,

   $$g = G + \pi,$$

   and decompose the nominal interest rate using the Fisher equation into the real interest rate R and inflation, π,

   $$r = R + \pi.$$

   Using the real interest rate and the real growth rate, determine the condition for government budget sustainability in the constant-r model. What are the implications for a government that hopes to sustain a large deficit by increasing inflation?

   Answer: The condition is the same in real terms: the budget is sustainable if $R < G$. To see why, replace r and g in Equation (3):

   $$\Delta b_{t+1}(b_t) = b_t \left(\frac{R + \pi - (G + \pi)}{1 + G + \pi} \right) + \frac{d}{1 + G + \pi} = b_t \left(\frac{R - G}{1 + G + \pi} \right) + \frac{d}{1 + G + \pi}$$

   If we were to draw the diagram for the constant-r model, we would see that $\Delta b_{t+1}(b_t)$ slopes down if $R < G$. Simply put, where we previously had $r - g$, inflation cancels out: $r - g = (R + \pi) - (G + \pi) = R - G$. Thus, permanently increasing the inflation rate does not make the government budget any more sustainable because the interest rate on debt will increase by the same amount as nominal GDP. The only way a government could credibly wipe away debt with inflation is if the inflation is a temporary surprise.

2. Derive the steady-state debt-to-GDP ratio in the constant-r model.

Answer: Using Equation (3), the steady-state debt-to-GDP ratio b^* solves

$$\Delta b_{t+1}(b^*) = b^* \left(\frac{r-g}{1+g}\right) + \frac{d}{1+g} = 0.$$

Solve for b^*:

$$\begin{aligned} b^*(r-g) + d &= 0 \\ b^*(r-g) &= -d \\ b^* &= \frac{-d}{r-g} \\ b^* &= \frac{d}{g-r} \end{aligned}$$

3. For the US government budget to be sustainable, it must decrease its primary deficit to 1.4% of GDP. Primary expenditures are 19.6% of GDP while tax revenue is 17.0% of GDP. To become sustainable, either taxes must increase to _____% of GDP or expenditure must decrease to _____% of GDP.

   Answer: To hit a primary deficit of 1.4% while keeping expenditure fixed at 19.6% of GDP, tax revenue must be $19.6 - 1.4 = 18.2\%$ of GDP. To hit a primary deficit of 1.4% while keeping tax revenue fixed at 17% of GDP, government expenditure must be $17.0 + 1.4 = 18.4\%$ of GDP.

4. Consider the international data in Table 3.

   (a) Japan has a very high debt-to-GDP ratio. Using the parameters in Table 3, compute the maximum primary deficit for Japan in 2016. Was that a sustainable budget? Why or why not?

   Answer: See Table 3. Yes, that budget was barely sustainable. Although growth was low and debt was high, interest rates were so low that it was sustainable.

   (b) Using the parameters in Table 3, compute the maximum primary deficit for China in 2016. Why is it so high?

Answer: China could theoretically sustain a very large deficit because growth was so high.

(c) In 2026, many countries, including Italy, have $r > g$. However, Italy's budget is still sustainable. How?

Answer: It runs a negative deficit, or budget surplus.

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